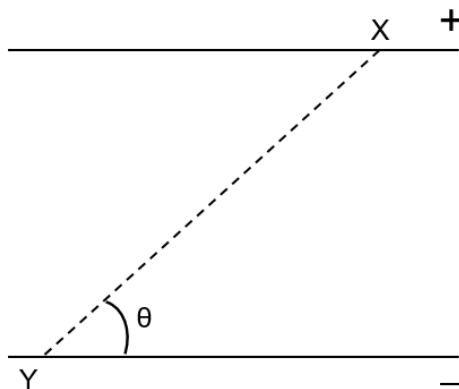


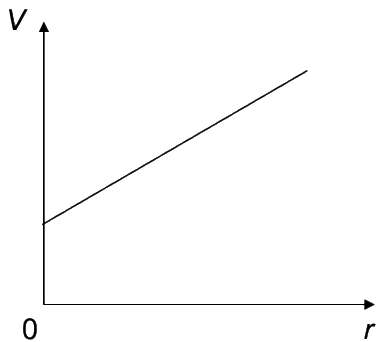
- 20** The figure below shows two horizontal parallel metal plates. The upper plate is maintained at a higher potential relative to the lower plate.



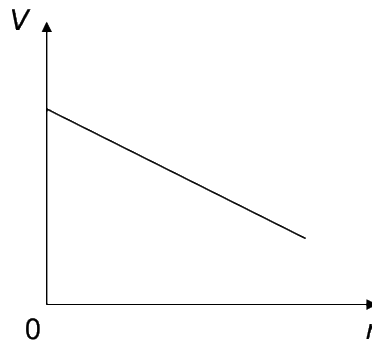
Line XY is inclined at an angle θ to the metal plates.

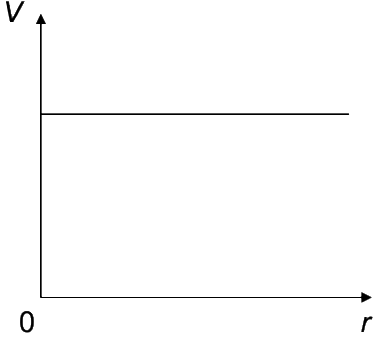
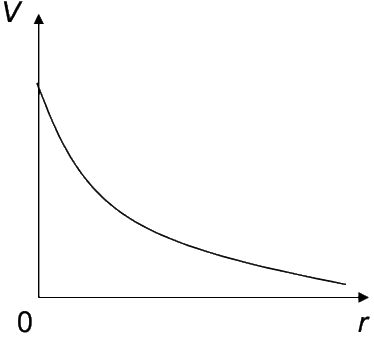
Which of the following graphs best shows the variation of electric potential V with distance r from X along the line XY?

A



B



C		D	
Answer: B Since θ and E are constant, V has a linear relationship to r, eliminating options C and D. E is the electric field strength between the plates. Furthermore, X is at a higher potential compared to Y. Therefore, option A is not correct. <u>Mathematical proof</u> Point P on XY is defined to be at a perpendicular distance $r\sin\theta$ from the upper plate. $E_P = \frac{\Delta V_{X \rightarrow P}}{r\sin\theta}$ $\Delta V_{X \rightarrow P} = E_P(r\sin\theta)$ $= r(E_P\sin\theta)$ Therefore, the graph's equation is given by $\begin{aligned} V &= V_P \\ &= V_X - \Delta V_{X \rightarrow P} \\ &= V_X - r(E_P\sin\theta) \end{aligned}$ Since E_P is the same at all points in a uniform field and $\sin\theta$ is a constant, the equation above can be simplified to $V = V_X - kr$ where k and V_X are constants.			

21 Two charges are fixed at a distance of 1.0 mm apart as shown below.